LINUX PROGRAMMING LAB

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A5IT07	PCC	-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The course should enable the students to:

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols

COURSE OUTCOMES

The course should enable the students to:

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Programs to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Communication using various protocols.
5. Connect remote systems using linux commands.

LIST OF EXPERIMENTS

WEEK - 1 LINUX COMMANDS

Linux Commands & Introduction of Shell

WEEK - 2 SHELL SCRIPTS

1. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
2. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.

WEEK - 3 FILES

a. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.

b. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file display the number of lines.

WEEK - 4 STANDARD I/O

a. Write a program that makes a copy of a file using standard I/O.

WEEK - 5 SYSTEM CALLS

a. Write a program that makes a copy of a file using standard I/O and system calls.

b. Implement in C the following UNIX commands using System calls.

i) cat ii) mv

WEEK - 6	DIRECTORY
a. Write a Program to list files in a directory. b. Write a program to list all of the files and its inode number in a directory	
WEEK - 7	**FORK FUNCTION**
Write a C program to create a child process and allow the parent to display “I am parent” and the child to display “I am child” on the screen.	
WEEK - 8	**PROCESS**
a. Write a C program to create a Zombie process. b. Write a C program that illustrates how an orphan is created.	
WEEK - 9	**PIPE**
Write a C program to implement pipe mechanism, where parent writes a message to a pipe and the child reads the message from the pipe and display the message.	
WEEK - 10	**FIFO**
Write C programs that illustrate communication between two unrelated processes using named pipe (FIFO).	
WEEK - 11	**MESSAGE QUEUES**
Write C program to implement message queues	
WEEK - 12	**SOCKET PROGRAMMING**
Write Client server program in C for connection oriented communication between server and client process using internet domain socket.	

TEXT BOOKS

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH,2006.

REFERENCE BOOKS

1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008
2. Unix Network Programming ,W.R.Stevens,PHI.
3. Linux System Programming, Robert Love, O‘Reilly, SPD.
4. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson
6. Education Internet and World Wide Web – How to program, Dietel and Nieto, Pearson

WEB LINKS

1. https://nptel.ac.in/courses/117/106/117106113/
2. https://www.linux.com/topic/desktop/understanding-linux-links/